

## Performance Testing Report

|               |                                                 |
|---------------|-------------------------------------------------|
| Date          | 24 June 2026                                    |
| Team ID       |                                                 |
| Project Name  | EcoTrack - Personal Carbon AI Footprint Tracker |
| Maximum Marks | 5 Marks                                         |

### Performance Testing Report:

Performance testing is critical to ensure your solution is fast, scalable, and stable under specific workloads. Document your testing strategy, the tools utilized, and the specific metrics measured against your target thresholds.

### Step 1: Test Scenarios & Tooling

| S.No | Test Type         | Scenario / API Endpoint Tested                                                                  | Tool(s) Used                   |
|------|-------------------|-------------------------------------------------------------------------------------------------|--------------------------------|
| 1    | Load Testing      | Simulated 500 concurrent users logging daily transport, diet, & energy habits via REST API.     | Apache JMeter & Postman Runner |
| 2    | Stress Testing    | Pushed Firebase Firestore & REST server with 5,000 requests/sec habit log writes.               | Locust Python & Artillery.io   |
| 3    | Spike Testing     | Sudden burst of 1,200 concurrent user requests to Groq LLaMA 3.3 AI Eco-Swaps API.              | k6 Cloud & Apache JMeter       |
| 4    | Endurance Testing | Continuous 24-hour baseline load test checking Flask API memory leakage and connection pooling. | Prometheus & Grafana Monitor   |

### Step 2: Key Performance Indicators (KPIs) & Results

| S.No | Metric (KPI)                     | Target Threshold | Actual Result     | Status |
|------|----------------------------------|------------------|-------------------|--------|
| 1    | <b>Average Response Time</b>     | < 200 ms         | 145 ms            | Pass   |
| 2    | <b>Peak Response Time</b>        | < 500 ms         | 320 ms            | Pass   |
| 3    | <b>Throughput (Requests/sec)</b> | > 1000 RPS       | 1450 RPS          | Pass   |
| 4    | <b>Error Rate</b>                | < 1.0%           | 0.12%             | Pass   |
| 5    | <b>CPU / Memory Utilization</b>  | < 80% under load | 62% CPU / 48% RAM | Pass   |